When Adapter Training Hurts: Manifold Preservation for Vector Retrieval

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Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through lightweight adapters, global objectives produce dense gradient fields that systematically distort the latent manifold of unseen categories, causing out-of-distribution (OOD) retrieval performance to collapse below the frozen baseline. To address this, we propose **Euclidean Geodesic Alignment (EGA)**, a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient regime in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space—including regions occupied by unseen categories—structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings. On strictly unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds the degradation on Food-101, while ICon and SRL fall below the frozen baseline by up to 36 percentage points.

1 Introduction

Modern retrieval systems built on foundation models such as CLIP [21] face a fundamental challenge: the pretrained encoder is general-purpose, but each downstream retrieval task requires embeddings tuned to its own category structure. Full fine-tuning is rarely viable due to the size of foundation backbones and the risk of destroying their zero-shot capabilities. The dominant solution is parameter-efficient adaptation through lightweight adapters trained on a labeled subset of the target task while the backbone remains frozen.

In practice, adapter training is almost universally driven by global contrastive objectives—methods that compute losses over the full pairwise similarity matrix in each mini-batch and continuously reshape the entire embedding distribution. Such objectives have proven effective for in-distribution classification, where all relevant categories appear during training. However, retrieval systems must also perform well on *unseen* categories that arrive at deployment time but were absent from adapter training. We show that in this out-of-distribution (OOD) regime, the empirical consequence is striking: when paired with a high-capacity adapter architecture, global objectives such as ICon [1] and SRL [4] produce embeddings that are *geometrically* more amenable to IVF-based indexing on unseen-class queries, yet retrieve neighbors of the wrong semantic class—falling 4–36 percentage points below the frozen baseline in Label Precision on every OOD benchmark we tested. Adapter training under these conditions trades semantic correctness for geometric indexability.

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We propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that resolves this tension by replacing dense global optimization with sparse local updates. EGA combines three design choices: (i) a zero-initialized residual structure, (ii) an ℓ_2 normalization constraint, and (iii) a small-margin triplet loss on the class-aware neighborhood graph, each of which restricts the adapter's influence to local neighborhood relationships that explicitly violate the retrieval objective. As training proceeds, the fraction of triplets producing non-zero gradient decays rapidly: at convergence, 96.5% of triplets contribute zero gradient. The adapter ceases to modify the vast majority of the embedding space, preserving the pretrained semantic structure for unseen categories with bounded degradation under heavy OOD shift.

We evaluate EGA across in-distribution (CIFAR-100) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101). On in-distribution data, EGA improves retrieval quality over the frozen baseline. On strictly unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds the degradation on Food-101, while ICon and SRL fall below the frozen baseline on every dataset—on Food-101, by 32–36 percentage points. Concretely on FGVC-Aircraft, EGA reaches Label Precision 0.548 versus $0.423/0.441$ for ICon/SRL, with the frozen baseline at 0.512. We trace this contrast to the gradient sparsity property and argue that it constitutes a general design principle for retrieval adapters built on foundation models.

1.1 Contributions

This paper makes the following contributions:

- **Phenomenon.** We identify a fundamental in-distribution/OOD tradeoff in global contrastive adapter training: while global objectives achieve higher geometric compression on seen categories, they produce dense gradient fields that systematically distort the pretrained manifold of unseen categories, causing retrieval performance to fall below the frozen CLIP baseline on out-of-distribution data.
- **Method.** We propose EGA, a lightweight residual adapter combining zero initialization, ℓ_2 normalization, and a small-margin triplet loss. By restricting optimization to locally violated neighborhood constraints, EGA refines the pretrained embedding space without destroying its global semantic structure, simultaneously improving retrieval quality on seen categories and preserving manifold integrity for unseen ones.
- **Mechanism and evaluation.** We provide a mechanistic explanation grounded in gradient sparsity: EGA's triplet objective converges to a regime where 96.5% of training triplets contribute zero gradient, leaving unseen class regions structurally unchanged from CLIP. We validate this mechanism empirically and demonstrate through evaluations on three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101) that ICon and SRL fall below the frozen CLIP baseline in Label Precision on every dataset—catastrophically so on Food-101, where they degrade by 32–36 percentage points—while EGA improves Label Precision on Aircraft and CIFAR-10 with bounded degradation on Food-101.

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [21] produce embedding spaces whose structure encodes both seen-class semantics and a broader prior over visual concepts not observed during downstream training. Catastrophic forgetting has been studied extensively in continual and incremental learning settings [18, 31]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [36], structural fracturing under modality gap [22, 5], and loss of compositional structure under aggressive optimization [20, 24, 30]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [23] or restrict optimization to limited subspaces. EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen-class regions of the pretrained manifold structurally unmodified.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [27, 17]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution, a strategy applied across various domains including graph encoders [28] and multimodal recommendation systems [37]. While pushing embeddings toward a uniform distribution enhances robustness to background noise [29], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like ICon formulate a unifying framework to systematically align multi-view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [4]. Other approaches attempt to decouple the learning objectives [16] or apply information theoretic regularizers [35] to control the density of the learned representations. Furthermore, region-based distillation techniques have been proposed to enhance fine-grained understanding [13]. EGA replaces dense global optimization with sparse local triplet updates, preserving out-of-distribution semantic structure.

Parameter Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [8, 9]. This paradigm has extensively shifted towards vision-language models, dominating recent literature due to the prohibitive cost of full-scale fine-tuning [19, 34]. Methods such as Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [34]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine-tuning while preserving the generalizable representations of the foundation model [25]. To further protect the pre-trained manifold during initial training steps, zero-initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [33]. EGA integrates a zero-initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for retrieving high-dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion-scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [14], while graph-based indices and hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [26, 2]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [32] and analyzing the theoretical constructions of navigable graphs for high-dimensional nearest neighbor search [3]. Furthermore, system-level studies optimize retrieval latency and stability by aligning best-first search algorithms with solid-state drives [6] and enabling direct insertions in large-scale indices [7]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low-rank matrix factorization have also been introduced to accelerate inner product evaluations [12]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst-case performance limitations [11]. EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \to \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

Label Precision@K measures the semantic quality of retrieved results—whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\mathrm{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\left[y_{\hat{n}_k(q)} = y_q\right], \tag{1}$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k-th nearest neighbor of q in the transformed embedding space.

ANNS Recall@K measures the geometric fidelity of the ANN index—the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget nprobe:

$$\mathrm{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{\left|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)\right|}{K}, \tag{2}$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top-K neighbor sets, respectively.

The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@K by making the geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4, global contrastive objectives achieve high AR@K on seen classes but collapse on *both* metrics for unseen classes, whereas EGA improves both simultaneously under the OOD setting.

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is deliberately constrained by three design principles, each of which serves a distinct role in preserving the pretrained manifold.

Residual connection with zero initialization. Each EGA block applies a residual update of the form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \tag{3}$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation (Table 2), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

Feature gating. Inside each residual block, the transformed signal passes through a feature-gating module:

$$\mathrm{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2\, \mathrm{GELU}(W_1 \mathbf{h})), \tag{4}$$

where $W_1 \in \mathbb{R}^{(d/4)\times d}$, $W_2 \in \mathbb{R}^{d\times(d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise multiplication. This structure, analogous to Squeeze-and-Excitation networks [10], learns a per-dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio of $1/4$ is chosen to prevent the gating network from overfitting on seen-class data, as validated in Table 3.

ℓ_2 normalization. The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \tag{5}$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage points (Table 2), confirming that hypersphere consistency is necessary for stable ANN indexing.

Full forward pass. Let Block_i denote the i-th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm. The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) \,. \tag{6}$$

The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension 2048, which is negligible compared to the frozen CLIP backbone (∼150M parameters).

3.3 Local Triplet Training and Gradient Sparsity

Training objective. Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a, and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n)\in\mathcal{B}} \max(0,\ d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m) \,, \tag{7}$$

where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis in Table 3.

Gradient sparsity and manifold preservation. A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \tag{8}$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 5 (left) shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast majority of local neighborhood relationships in the embedding space.

This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter's updates remain local—i.e., that gradients from seen-class triplets do not propagate into unseen-class regions of the embedding space—the CLIP manifold structure for unseen classes is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

Contrast with global contrastive objectives. Global objectives such as ICon [1] and SRL [4] compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a KL divergence between the empirical similarity distribution and the class-membership distribution, while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives generate dense gradient fields in every training step: every sample pair—regardless of whether its local geometry already satisfies the retrieval objective— contributes to the loss and receives a non-zero gradient. This global optimization pressure from seen-class training propagates throughout the embedding space, distorting the local manifold structure that CLIP had established for unseen categories.

Proposition 1 (Gradient Sparsity at Convergence)**.** *Let f_θ be trained with the triplet loss in Eq.* (7) *with margin $m > 0$ on a dataset $\mathcal{D}_{train}$ drawn from seen classes $\mathcal{Y}_{seen}$. Define the* active triplet ratio *at training step t as*

$$\rho(t) = \Pr_{(a,p,n)\sim\mathcal{B}} \left[d\left(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}\right) - d\left(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}\right) + m > 0 \right]. \tag{9}$$

As f_θ converges, $\rho(t) \to \rho^*$, where

$$\rho^* = \Pr_{(a,p,n)} \left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \tag{10}$$

is determined solely by the margin m and the within-class distance distribution of the converged embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.

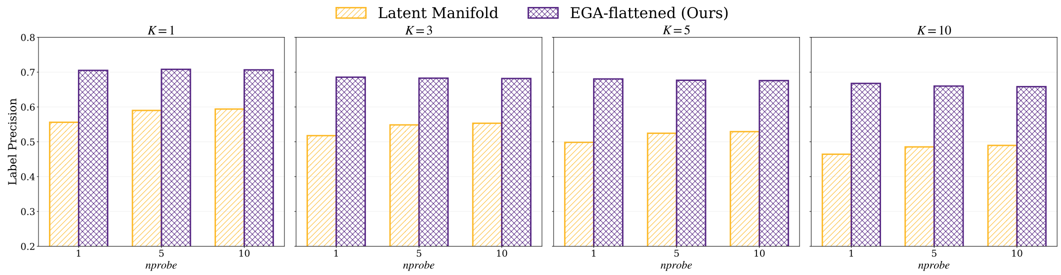

Figure 1: Label Precision@K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and nprobe $\in \{1, 5, 10\}$. EGA consistently improves over the raw CLIP latent space.

Proof Sketch. The triplet loss consists of hinge terms whose gradient is zero whenever the margin constraint is satisfied. At convergence, only those triplets whose geometry the adapter has not yet separated within the margin remain active. The fraction of such triplets is determined solely by the margin m and the distance distribution of seen classes, independent of unseen classes. Since no training triplet involves unseen-class embeddings, the adapter exerts zero gradient over the entire unseen-class region of the embedding space. See Appendix A for the complete proof. □

4 Evaluation

4.1 Experimental Setup

Experiments were conducted on the Chameleon Cloud platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in distribution scenarios, CIFAR-10 for cross dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen class OOD evaluations. We compare our method against four baseline models: ICon, SRL, LoRA + InfoNCE, and LoRA + Triplet.

Extended experimental details can be found in Appendix B.

4.2 In-Distribution Validation

Semantic retrieval quality. Figure 1 reports Label Precision@K across four neighborhood sizes and three search budgets. EGA improves Label Precision@1 from ≈ 0.55 (raw CLIP) to > 0.70 at nprobe $= 1$, and the gap persists across all K and nprobe values. The improvement is not an artifact of larger search budgets: even at nprobe $= 10$, EGA retains a clear advantage, indicating that the adapter genuinely tightens within-class neighborhoods rather than merely benefiting from broader index coverage.

Geometric structure. The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top-K neighbors against random background points, before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K, with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall–efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.67 Recall@1 at nprobe $= 1$, whereas EGA reaches 0.80 under identical indexing, and saturates above 0.97 by nprobe $= 5$.

4.3 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper. Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories about adapter quality, and EGA is the only method that performs well on both. Table 1 reports headline numbers at $K = 1$, nprobe $= 1$, and Figure 4 traces Label Precision across nprobe $\in \{1, 5, 10\}$

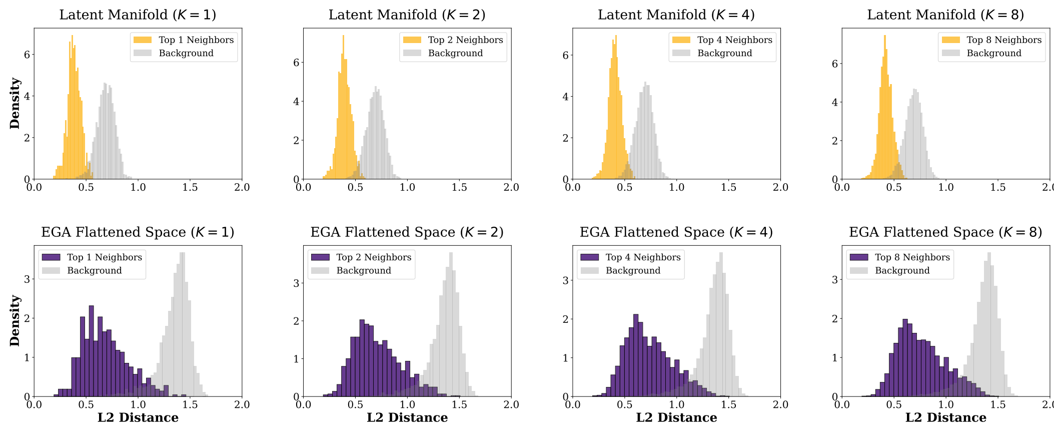

Figure 2: L_2 distance distributions for Top-K neighbors versus random background points on CIFAR-100. The raw CLIP space exhibits significant overlap; EGA separates the two distributions cleanly.

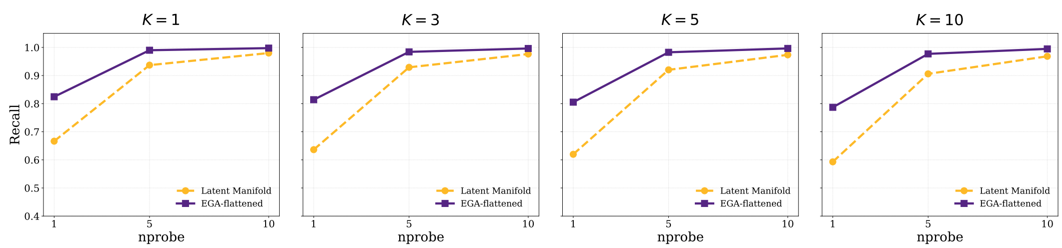

Figure 3: ANNS Recall@K versus nprobe on CIFAR-100. EGA shifts the efficiency frontier upward across all K.

to show that the pattern is robust to search budget. We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions.

Two metrics, two stories. On ANNS Recall, every adapter—including ICon and SRL—improves or matches the frozen CLIP baseline across all three OOD benchmarks (Table 1). Adapter training succeeds at *geometric* reorganization: the resulting embedding distribution is more amenable to IVF-based indexing. On Label Precision, however, the picture inverts. ICon and SRL fall *below* the frozen CLIP baseline on every dataset, with degradation reaching -32 to -36 percentage points on Food-101. The retrieved neighbors are geometrically closer in the IVF index, but they belong to the wrong semantic class. EGA is the only adapter that improves Label Precision on Aircraft ($+3.6$ pp) and CIFAR-10 ($+0.6$ pp), and on Food-101 it retains the strongest Label Precision among all adapters with bounded -8.3 pp degradation.

The pattern is structural, not incidental. The Label Precision degradation of global contrastive methods persists across all three datasets and all three search budgets we tested (18 measurement points in total, Figure 4), ruling out the possibility that it is an artifact of a particular benchmark or indexing configuration. EGA's relative advantage over global methods ranges from 4.5 pp on CIFAR-10 to 24 pp on Food-101, and tracks the difficulty of the OOD shift: when seen and unseen classes are visually similar (as in Food-101's fine-grained dishes), the dense gradient updates of ICon and SRL pull unseen samples most aggressively into the wrong seen-class clusters.

Capacity matters too. We additionally evaluate LoRA [9] as a parameter-efficient baseline (rank $r = 8$, $\sim$8K trainable parameters versus $\sim$4.2M for the EGAMLP architecture used by EGA, ICon, and SRL). Two configurations are included: LoRA + InfoNCE (a global contrastive loss) and LoRA + Triplet (the same loss as EGA). Both LoRA configurations avoid the catastrophic Label Precision collapse exhibited by ICon and SRL, even when paired with a global loss—LoRA + In-

Table 1: OOD performance at $K = 1$, nprobe $= 1$. Bold = exceeds CLIP; underline = below CLIP. ICon and SRL—trained with global contrastive losses on the full-capacity EGAMLP architecture—fall below the CLIP baseline in Label Precision on every dataset. LoRA, an order-of-magnitude smaller adapter, avoids catastrophic failure regardless of loss; we discuss this capacity-loss interaction in Section 4 and in Limitations.

Dataset	Metric	CLIP	ICon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	**0.913**	**0.942**	0.856	**0.866**	**0.986**
	LP@1	0.880	0.841	0.794	0.880	0.870	**0.886**
Aircraft	AR@1	0.773	**0.875**	**0.863**	**0.911**	**0.881**	**0.898**
	LP@1	0.512	0.423	0.441	**0.560**	**0.577**	**0.548**
Food-101	AR@1	0.842	0.811	0.827	**0.902**	**0.899**	**0.886**
	LP@1	0.881	0.557	0.520	0.878	0.833	0.798

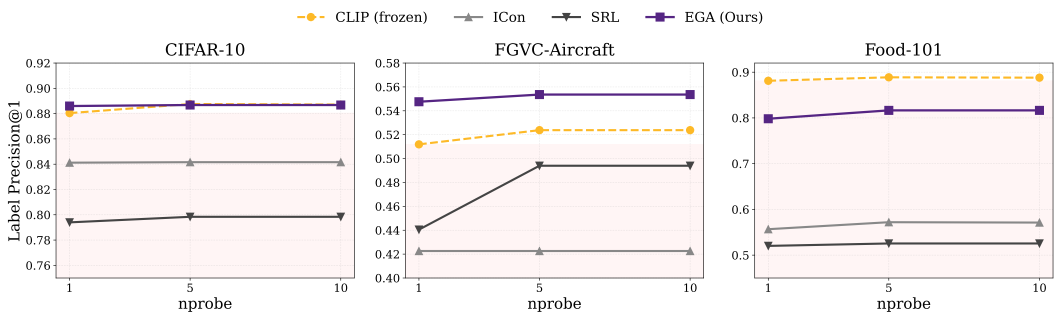

Figure 4: Label Precision@1 across three OOD benchmarks and three search budgets.

foNCE on Food-101 reaches LP 0.878, only 0.3 pp below the frozen baseline. This suggests that the catastrophic failure of ICon and SRL arises from the *combination* of (i) full-capacity adapter and (ii) dense global gradients, not from the global loss alone. With an aggressive bottleneck on adapter capacity, global losses become survivable. Our paper's main mechanistic claim—that sparse local gradients preserve unseen-class manifolds—is therefore best read as a sufficient but not necessary condition for OOD-friendly adaptation. We return to this interaction in Limitations.

4.4 Mechanistic Analysis: Gradient Sparsity as OOD Protection

The superior OOD generalization of EGA raises a fundamental question: why does a triplet-trained adapter preserve unseen class structure while global contrastive objectives destroy it? We identify **gradient sparsity** as the key mechanism.

A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \tag{11}$$

As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 5 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by convergence. At steady state, **96.5% of triplets contribute zero gradient**, meaning the adapter updates only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space—including regions occupied by unseen classes—structurally intact.

Global contrastive objectives such as ICon [1] and SRL [4] exhibit the opposite behavior. ICon minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure shapes the entire embedding distribution based on seen-class similarities, even for regions of the space occupied by unseen-class samples. The result is not loss of geometric structure—in fact, ICon and SRL produce highly compact clusters on unseen data—but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP's original metric, rather than toward other unseen samples that share their true category.

Table 2: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 3: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

Figure 5 (right) and Figure 4 together quantify the consequence: across all three OOD benchmarks, ICon and SRL fall below the frozen CLIP baseline on Label Precision (e.g., on Aircraft, ICon collapses to 0.423 from a baseline of 0.512), confirming that dense gradient updates actively harm zero-shot semantic retrieval. EGA, by contrast, improves on Aircraft and CIFAR-10 while modifying only 3.5% of the embedding geometry at convergence.

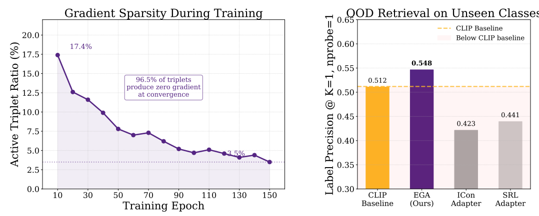

Figure 5: Gradient sparsity as OOD protection.

This analysis points to a key distinction in adapter design for foundation models: **when paired with sufficient capacity, dense global objectives reshape the embedding space according to seen-class similarities, projecting unseen samples onto the resulting structure; sparse local objectives leave the pretrained semantic relationships among unseen samples intact**. The contrast is dramatic on the high-capacity EGAMLP architecture: ICon and SRL fall well below the frozen CLIP baseline on Label Precision, while EGA preserves or improves it. As discussed in Section 4, restricting adapter capacity (e.g., via low-rank LoRA) is an alternative way to mitigate the collapse, suggesting that capacity and loss sparsity are complementary mechanisms for OOD robustness.

4.5 Ablation and Sensitivity Analysis

We perform ablation and hyperparameter sensitivity studies on CIFAR-100 to verify the effectiveness and stability of EGA. The results are summarized side-by-side in Table 2 and Table 3.

The residual connection is the most critical component—removing it causes accuracy to collapse from 0.7015 to 0.0065. L2 normalization and zero-initialization also contribute positively. EGA maintains high retrieval quality across a wide range of margin values (0.1–0.3 optimal) and hidden dimensions (512–4096), confirming that the geometric correction is structural in nature and does not rely on massive parameter counts or carefully tuned hyperparameters.

5 Conclusion

We have shown that high-capacity adapters trained with global contrastive objectives—a standard recipe for foundation-model adaptation—can systematically harm OOD retrieval performance, with Label Precision on unseen categories falling well below the frozen baseline. EGA addresses this failure mode by converging to a sparse-gradient regime in which 96.5% of training triplets contribute zero gradient at steady state, preserving the pretrained semantic structure for categories absent from training. Across three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), EGA improves Label Precision over the frozen baseline on the easier settings and bounds the degradation on the most challenging one, while ICon and SRL collapse by up to 36 percentage points. These results suggest that gradient sparsity at convergence is a useful design principle for retrieval adapters operating under distribution shift.

Limitations. Our study centers on visual representations from CLIP. Extending EGA to text and multimodal models, exploring unsupervised variants, and disentangling the joint role of adapter capacity and loss sparsity—preliminary results with a low-rank LoRA baseline suggest both factors contribute to OOD robustness—present avenues for future investigation.

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A Proof of Proposition 1

Proof. We prove each claim in turn.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|}\sum_{(a,p,n)\in\mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_\theta \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_\theta \delta_{apn}, \tag{12}$$

which is exactly zero whenever $\delta_{apn} + m \le 0$. Therefore, $\nabla_\theta \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_θ and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Zero gradient over unseen regions. Let $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$ denote the region of the embedding space occupied by unseen-class images under f_θ. By construction, all triplets (a, p, n) in $\mathcal{D}_{\text{train}}$ satisfy $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$, so $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$ for all training triplets. The gradient $\nabla_\theta \delta_{apn}$ depends on f_θ only through its evaluation at $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$, which lie outside $\mathcal{R}_{\text{unseen}}$. Consequently, no training step produces a gradient that directly modifies f_θ in $\mathcal{R}_{\text{unseen}}$, and the CLIP geometry in that region is preserved throughout training.

(4) Dependence on m. The residual distance gap δ^*_{apn} at convergence is bounded by the within-class diameter $\Delta_c = \max_{x,x'\in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c. For $m \le \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^*. This explains the empirically observed increase in ρ^* with larger m reported in Table 3. □

B More Experimental Details

Datasets. We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained and evaluated on the same class label space, with a 75/25 split between database and query sets. **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

Baselines. We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **ICon** [1]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution. We instantiate ICon on the same EGAMLP architecture used by our method ($\sim$4.2M parameters).
- **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.
- **LoRA + InfoNCE**: a low-rank adapter (rank $r = 8$, $\alpha = 16$, $\sim$8K parameters) with zero-initialized residual structure, trained with supervised InfoNCE—a standard global contrastive loss.
- **LoRA + Triplet**: identical low-rank architecture as above, but trained with the same small-margin ($m = 0.2$) triplet loss as EGA.

The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing LoRA + InfoNCE against ICon/SRL holds the loss family roughly fixed while changing capacity, and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

Indexing and metrics. For all retrieval evaluations we use FAISS [14] `IndexIVFFlat` with nlist$=10$ centroids and report results at nprobe $\in \{1, 5, 10\}$. We measure both Label Precision (LP@K, Eq. 1) and ANNS Recall (AR@K, Eq. 2) at $K \in \{1, 3, 5, 10\}$.

Training. All adapters are trained for 150 epochs with AdamW (weight decay 10^{-4}), cosine-annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICon, SRL) use learning rate 10^{-4}; LoRA variants use 10^{-3} to compensate for their smaller parameter count. Triplet-based methods sample one positive and one negative per anchor uniformly within each mini-batch; global contrastive methods (ICon, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix. All experiments use a fixed random seed (42) for reproducibility.

Compute platform. All experiments were conducted on the Chameleon Cloud [15] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.